

CS5340

Uncertainty Modeling in AI

Lecture 3: Markov Random Fields (Undirected Graphical Models)

Assoc. Prof. Lee Gim Hee

AY 2026/27

Semester 1

Course Schedule

Week	Date	Topic	Remarks
1	10 Aug	National Day Holiday	No Lecture
2	17 Aug	Introduction to probabilistic reasoning	Assignment 0: Python Numpy Tutorial (Ungraded)
	22 Aug	Bayesian networks (Directed graphical models)	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
3	24 Aug	Markov random Fields (Undirected graphical models)	
4	31 Aug	Variable elimination and belief propagation	Assignment 1: Belief propagation and maximal probability (15%) Assignment 1: Due (13 Sep, 2359 HRS)
	05 Sept	Factor graph and the junction tree algorithm	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
5	07 Sep	I will be traveling	No Lecture
6	14 Sep	Parameter learning with complete data	Assignment 2: Junction tree and parameter learning (15%) Assignment 2: Due (27 Sep, 2359 HRS)
-	21 Sep	Recess week	No Lecture
7	28 Sep	Mixture models and the EM algorithm	
8	05 Oct	Hidden Markov Models (HMM)	Assignment 3: Hidden Markov model (15%) Assignment 3: Due (18 Oct, 2359 HRS)
9	12 Oct	Monte Carlo inference (Sampling)	
10	19 Oct	Variational inference	Assignment 4: MCMC Sampling (15%) Assignment 4: Due (01 Nov, 2359 HRS)
11	26 Oct	Variational Autoencoder and Diffusion Models	
12	02 Nov	Graph-cut and alpha expansion	
13	09 Nov	Diwali/Deepavali	No Lecture

Final Exam: 21 Nov 2026

Acknowledgements

- A lot of slides and content of this lecture are adopted from:
 1. "Machine learning - a probabilistic approach", Kevin Murphy (Chapter 19)
 2. "Probabilistic graphical models", Koller and Friedman (Chapter 4)
 3. "An introduction to probabilistic graphical models", Michael I. Jordan, 2002 (Section 2.2)
<http://people.eecs.berkeley.edu/~jordan/prelims/chapter2.pdf>
 4. "Pattern recognition and machine learning", Christopher Bishop (Chapter 8, Section 8.3).
 5. <http://www.cs.cmu.edu/~epxing/Class/10708/lectures/lecture3-MRFrepresentation.pdf>, Eric Xing

Learning Outcomes

- Students should be able to:
 1. Derive the five rules of conditional independence: **Symmetry**, **Decomposition**, **Weak Union**, **Contraction**, **Intersection**.
 2. Use **I-Map** to explain the correctness of graphical models for conditional independence.
 3. Explain the concepts of **Markov properties (global, local and pairwise)** and use it to find all conditional independences in an UGM.
 4. Use **clique potential functions** to parameterize a Markov Random Field, i.e. to represent the joint distribution with clique potential functions.
 5. Describe the differences and similarities between a **Markov Random Field** and **Conditional Random Field**.

Rules of Conditional Independence

- Recall we learned **conditional independence** in the last lecture:

$$X_A \perp X_C \mid X_B \Rightarrow \begin{cases} p(x_A, x_C \mid x_B) = p(x_A \mid x_B)p(x_C \mid x_B) \\ p(x_A \mid x_B, x_C) = p(x_A \mid x_B), \end{cases}$$

$$\forall X_B: p(x_B) > 0.$$

- Five additional rules** of conditional independence can be further derived.

Rules of Conditional Independence

1. **Symmetry**: $X \perp Y \mid Z \Rightarrow Y \perp X \mid Z$

Proof:

Using product rule, we have: $p(Y \mid X, Z) = \frac{p(X, Y \mid Z)}{p(X \mid Z)}$

$$\forall p(X \mid Z) > 0$$

Applying $X \perp Y \mid Z$ on $p(X, Y \mid Z)$, we get:

$$p(Y \mid X, Z) = \frac{p(X, Y \mid Z)}{p(X \mid Z)} = \frac{p(X \mid Z)p(Y \mid Z)}{p(X \mid Z)} = p(Y \mid Z)$$

Thus, Y is conditionally independent of X given Z , i.e. $Y \perp X \mid Z$.



Rules of Conditional Independence

2. Decomposition:

$$X \perp \{Y, W\} \mid Z \Rightarrow X \perp Y \mid Z \text{ and } X \perp W \mid Z$$

Proof:

$$X \perp \{Y, W\} \mid Z \Rightarrow p(X, Y, W \mid Z) = p(X \mid Z)p(Y, W \mid Z)$$

$$p(X, Y \mid Z) = \int p(X, Y, W \mid Z) dW = \int p(X \mid Z)p(Y, W \mid Z) dW$$

$$= p(X \mid Z)p(Y \mid Z) \Rightarrow X \perp Y \mid Z$$

$X \perp W \mid Z$ can be shown in similar way.

□

Rules of Conditional Independence

3. **Weak Union:**

$$X \perp \{Y, W\} \mid Z \Rightarrow X \perp Y \mid \{W, Z\} \quad \text{and} \quad X \perp W \mid \{Y, Z\}$$

Proof:

$$p(X, Y \mid W, Z) = \frac{p(X, Y, W \mid Z)}{p(W \mid Z)} \quad \forall p(W \mid Z) > 0 \quad (\text{product rule})$$

$$= \frac{p(X \mid Z)p(Y, W \mid Z)}{p(W \mid Z)} \quad \begin{array}{l} X \perp \{Y, W\} \mid Z \\ \Rightarrow p(X, Y, W \mid Z) = p(X \mid Z)p(Y, W \mid Z) \end{array}$$

$$= p(X \mid Z)p(Y \mid W, Z) \quad (\text{product rule})$$

$$= p(X \mid W, Z)p(Y \mid W, Z) \quad \text{Decomposition: } X \perp \{Y, W\} \mid Z \Rightarrow X \perp W \mid Z$$

Therefore, we have $X \perp \{Y, W\} \mid Z \Rightarrow X \perp Y \mid \{W, Z\}$;

$X \perp \{Y, W\} \mid Z \Rightarrow X \perp W \mid \{Y, Z\}$ can be proven in the same way.

□

Rules of Conditional Independence

4. Contraction:

$$X \perp Y \mid \{W, Z\} \text{ and } X \perp W \mid Z \Rightarrow X \perp \{Y, W\} \mid Z$$

Proof:

$$X \perp Y \mid \{W, Z\} \text{ and } X \perp W \mid Z$$

$$\Rightarrow p(X \mid Y, W, Z) = p(X \mid W, Z) = p(X \mid Z)$$

$$\Rightarrow X \perp \{Y, W\} \mid Z$$



Rules of Conditional Independence

5. Intersection:

$$X \perp Y \mid \{W, Z\} \text{ and } X \perp W \mid \{Y, Z\} \Rightarrow X \perp \{Y, W\} \mid Z$$

Proof:

$$X \perp Y \mid \{W, Z\} \Rightarrow p(X \mid W, Y, Z) = p(X \mid W, Z)$$

$$X \perp W \mid \{Y, Z\} \Rightarrow p(X \mid W, Y, Z) = p(X \mid Y, Z)$$

$$\Rightarrow p(X \mid W, Z) = p(X \mid Y, Z)$$

Rules of Conditional Independence

Proof (cont.):

$$p(X | Z) = \int p(X, W | Z) dW \quad \text{(marginalization)}$$

$$= \int p(X | W, Z) p(W | Z) dW \quad \text{(product rule)}$$

$$= p(X | Y, Z) \int p(W | Z) dW$$

$X \perp Y | \{W, Z\}$
 $X \perp W | \{Y, Z\}$

$$= p(X | Y, Z)$$

$$\begin{aligned} \Rightarrow p(X | W, Y, Z) &= p(X | W, Z) = p(X | Y, Z) \\ &= p(X | Z) \end{aligned}$$

$$\Rightarrow X \perp \{Y, W\} | Z$$

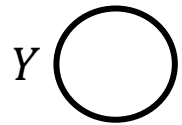
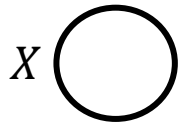
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Independence-Map

- Also known as the **I-Map**.
- The I-Map of a **joint distribution** $p(x_1, \dots, x_N)$, often written as $I(p)$ represents **all independencies** in $p(x_1, \dots, x_N)$.
- Similarly, the I-Map of a **directed/undirected graph** $\mathcal{G}$, i.e., $I(\mathcal{G})$ represents **all independencies** encoded in $\mathcal{G}$.
- $\mathcal{G}$ is a valid representation of p if $I(\mathcal{G}) \subseteq I(p)$.

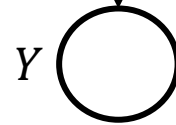
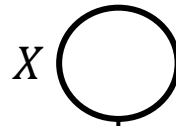
Independence-Map

Example 1:



$\mathcal{G}_0$ encodes $X \perp Y$
or

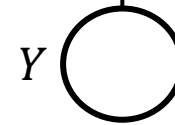
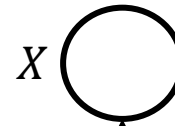
$$I(\mathcal{G}_0) = X \perp Y$$



$\mathcal{G}_1$ encodes no
independence

or

$$I(\mathcal{G}_1) = \emptyset$$



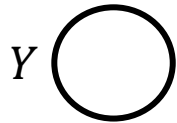
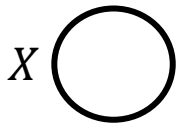
$\mathcal{G}_2$ encodes no
independence

or

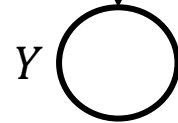
$$I(\mathcal{G}_2) = \emptyset$$

Independence-Map

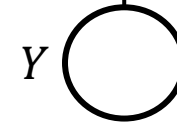
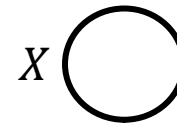
Example 1: Are $\mathcal{G}_0$, $\mathcal{G}_1$ and $\mathcal{G}_2$ valid representations of $p(X, Y)$?



$$I(\mathcal{G}_0) = X \perp Y$$



$$I(\mathcal{G}_1) = \emptyset$$



$$I(\mathcal{G}_2) = \emptyset$$

X	Y	$P(X, Y)$
0	0	0.08
0	1	0.32
1	0	0.12
1	1	0.48

Check for **mutual independence**:

$$p(X, Y) = p(X)p(Y) \text{ if}$$

$$p(Y | X) = p(Y) \text{ and}$$

$$p(X | Y) = p(X)$$

Independence-Map

Example 1:

X	$P(X) = \sum_Y p(X, Y)$
0	$p(X = 0, Y = 0) + p(X = 0, Y = 1)$ $= 0.08 + 0.32 = 0.40$
1	$p(X = 1, Y = 0) + p(X = 1, Y = 1)$ $= 0.12 + 0.48 = 0.60$

$\Updownarrow P(X | Y) = P(X)$

X	Y	$P(X Y) = p(X, Y)/p(Y)$
0	0	$\frac{p(X = 0, Y = 0)}{p(Y = 0)} = \frac{0.08}{0.2} = 0.4$
0	1	$\frac{p(X = 0, Y = 1)}{p(Y = 1)} = \frac{0.32}{0.8} = 0.4$
1	0	$\frac{p(X = 1, Y = 0)}{p(Y = 0)} = \frac{0.12}{0.2} = 0.6$
1	1	$\frac{p(X = 1, Y = 1)}{p(Y = 1)} = \frac{0.48}{0.8} = 0.6$

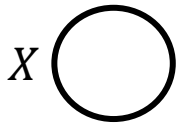
Y	$P(Y) = \sum_X p(X, Y)$
0	$p(X = 0, Y = 0) + p(X = 1, Y = 0)$ $= 0.08 + 0.12 = 0.20$
1	$p(X = 0, Y = 1) + p(X = 1, Y = 1)$ $= 0.32 + 0.48 = 0.80$

$\Updownarrow P(Y | X) = P(Y)$

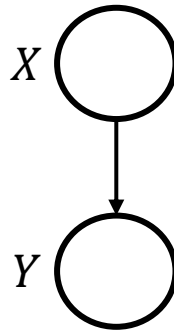
X	Y	$P(Y X) = p(X, Y)/p(X)$
0	0	$\frac{p(X = 0, Y = 0)}{p(X = 0)} = \frac{0.08}{0.4} = 0.2$
0	1	$\frac{p(X = 0, Y = 1)}{p(X = 0)} = \frac{0.32}{0.4} = 0.8$
1	0	$\frac{p(X = 1, Y = 0)}{p(X = 1)} = \frac{0.12}{0.6} = 0.2$
1	1	$\frac{p(X = 1, Y = 1)}{p(X = 1)} = \frac{0.48}{0.6} = 0.8$

Independence-Map

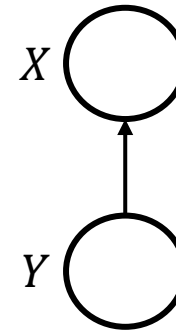
Example 1:



$$I(\mathcal{G}_0) = X \perp Y$$



$$I(\mathcal{G}_1) = \emptyset$$



$$I(\mathcal{G}_2) = \emptyset$$

$\mathcal{G}_0, \mathcal{G}_1, \mathcal{G}_2$ are **I-Maps** of $p(X, Y)$!

X	Y	$P(X, Y)$
0	0	0.08
0	1	0.32
1	0	0.12
1	1	0.48

There's **mutual independence**:

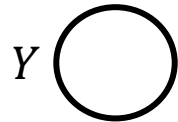
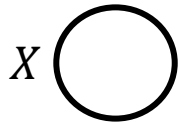
$$p(X, Y) = p(X)p(Y) \text{ since}$$

$$p(Y | X) = p(Y) \text{ and}$$

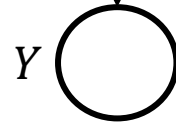
$$p(X | Y) = p(X)$$

Independence-Map

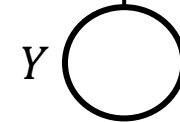
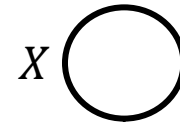
Example 2 (Exercise):



$$I(\mathcal{G}_0) = X \perp Y$$



$$I(\mathcal{G}_1) = \emptyset$$



$$I(\mathcal{G}_2) = \emptyset$$

X	Y	$P(X, Y)$
0	0	0.4
0	1	0.3
1	0	0.2
1	1	0.1

Show that:

$\mathcal{G}_0$ is not an I-Map of $P(X, Y)$

$\mathcal{G}_1, \mathcal{G}_2$ are I-Maps of $P(X, Y)$

From I-Map to Factorization

- If $\mathcal{G}$ is a valid representation of P , then

$$p(X_1, \dots, X_N) = \prod_{c \in \{1, \dots, N\}} f_c(X_c)$$

- $\prod_{c \in \{1, \dots, N\}} f_c(X_c)$ is the factorization according to $\mathcal{G}$.
- For example, the parent-child relationship in Bayesian networks:

$$p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i | x_{\pi_i})$$

- Thus, we can go from **graphs to factorization** of P

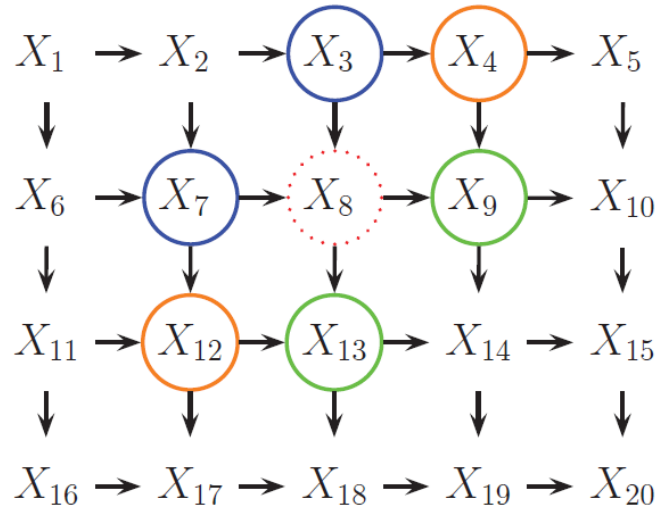
Why Undirected Graphical Models?

- We discussed the **Directed Graphical Models** (DGMs) or Bayesian Networks in the last lecture.
- However, for some domains, the requirement for a directed edge is **rather awkward**.

Why Undirected Graphical Models?

Example:

Causal MRF



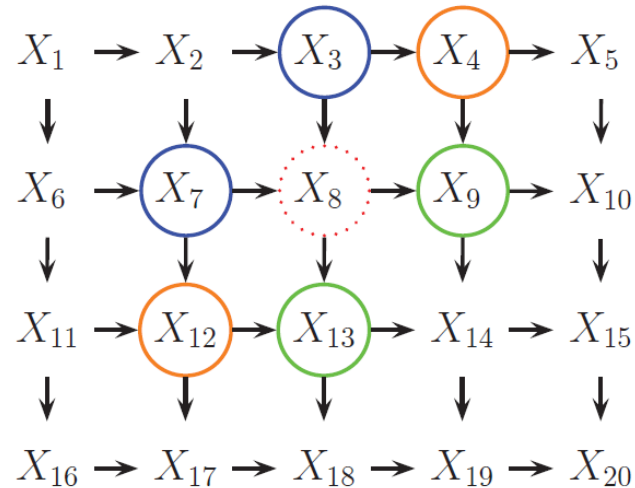
- Modeling a 2D image where the intensity of neighboring pixels are correlated.
- We can create a DAG model with a 2d lattice topology known as a **causal MRF** or a **Markov mesh**.

Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Why Undirected Graphical Models?

Example:

Causal MRF



- However, its conditional independence properties are rather **unnatural**.
- The **Markov blanket** of the node X_8 in the middle is the other colored nodes (3, 4, 7, 9, 12 and 13) rather than just its 4 nearest neighbors as one might expect.

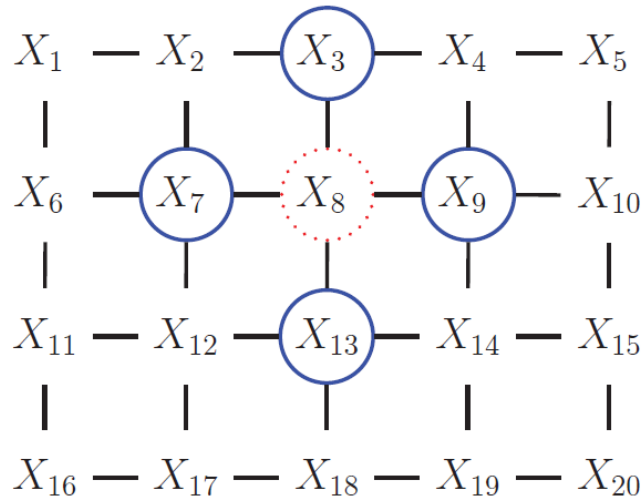
Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Why Undirected Graphical Models?

- An alternative is to use an **Undirected Graphical model (UGM)**, also called a **Markov Random Field (MRF)** or **Markov network**.
- Formally, an UGM is a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where:
 - $\mathcal{V}$ is a set of **nodes** that are in one-to-one correspondence with a set of random variables.
 - $\mathcal{E}$ is a set of **undirected** edges.
- No edge orientations, hence **more natural** for some problems such as **image analysis** and **spatial statistics**.

Why Undirected Graphical Models?

Example:



- We use an **undirected 2d lattice** to model a 2D image where the intensity of neighboring pixels are correlated.
- Now the **Markov blanket** of each node is just its nearest neighbors (more on Markov blanket for UGMs later).

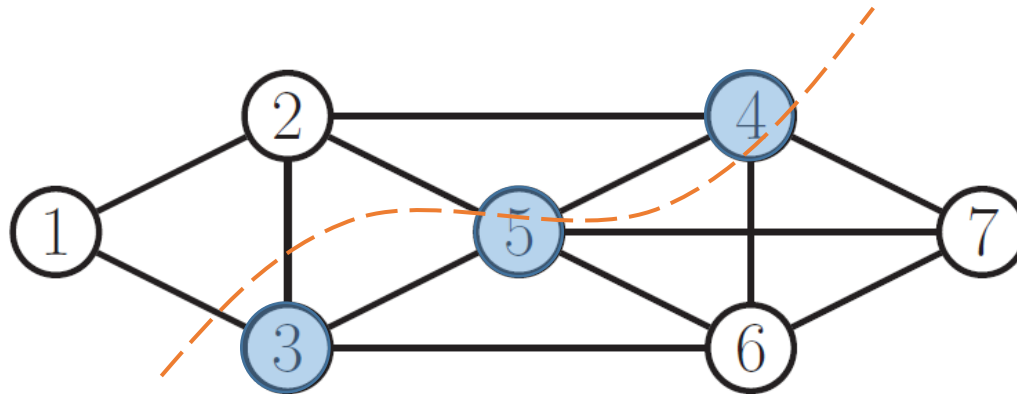
Image Source: “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

Conditional Independence

1. Global Markov Property

- Given the sets of nodes A, B and C , $X_A \perp X_B \mid X_C$ if and only if C separates A from B in the graph $\mathcal{G}$.
- This means that there are **no paths** connecting any node in A to any node in B when we remove all nodes in C .

Example:



$$\{X_1, X_2\} \perp \{X_6, X_7\} \mid \{X_3, X_4, X_5\}$$

Image Source: Modified from “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

Conditional Independence

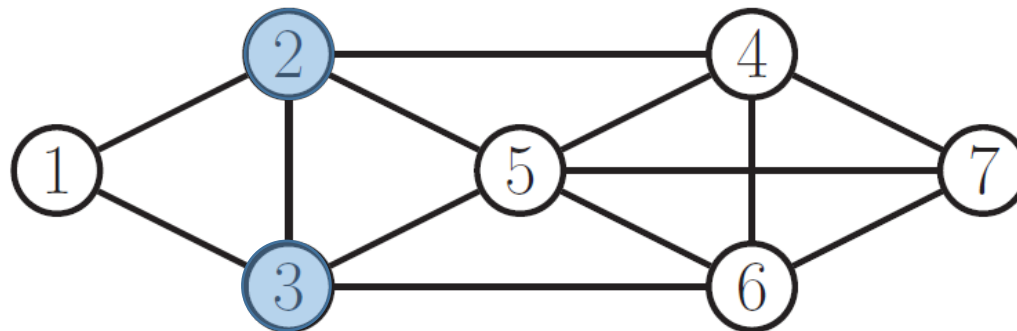
2. Local Markov Property

- The set of nodes that renders a node X_s conditionally independent of all the other nodes in $\mathcal{G}$:

$$X_s \perp \mathcal{V} \setminus \{\text{mb}(X_s), X_s\} \mid \text{mb}(X_s)$$

- where $\text{mb}(X_s)$ is called the **Markov blanket** of X_s .

Example:



$$\text{mb}(X_1) = \{X_2, X_3\}, \text{ i.e. } X_1 \perp \{X_4, X_5, X_6, X_7\} \mid \{X_2, X_3\}$$

Image Source: Modified from “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

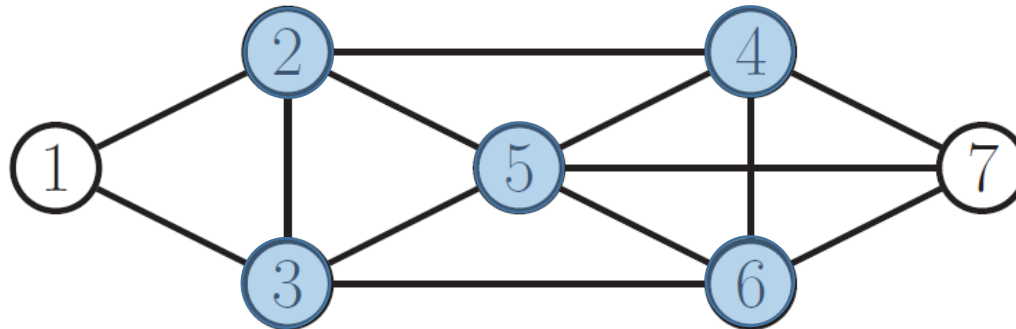
Conditional Independence

3. Pairwise Markov Property

- Two nodes X_s and X_t are conditionally independent given the rest if there is **no direct edge** between them:

$$X_s \perp X_t \mid \mathcal{V} \setminus \{X_s, X_t\}, \text{ where } \mathcal{E}_{st} = \emptyset$$

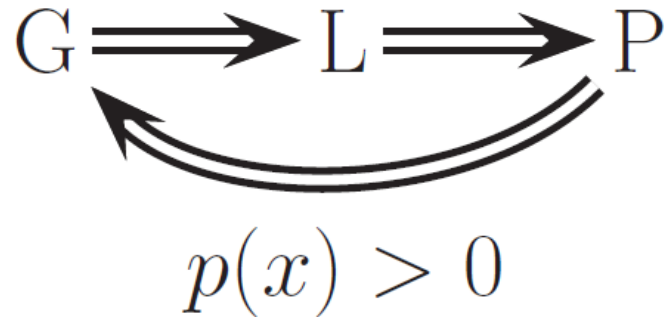
Example:



$$X_1 \perp X_7 \mid \{X_2, X_3, X_4, X_5, X_6\}$$

Image Source: Modified from “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

Conditional Independence

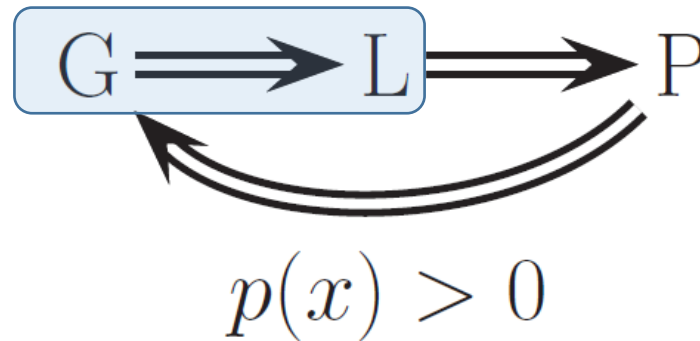


- Obvious that global Markov implies local Markov which implies pairwise Markov.
- What is less obvious, but true (assuming $p(\mathbf{x}) > 0$ for all $\mathbf{x}$), is that **pairwise Markov implies global Markov**.

Image Source: “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

Conditional Independence

Global Markov implies local Markov:



Proof Sketch:

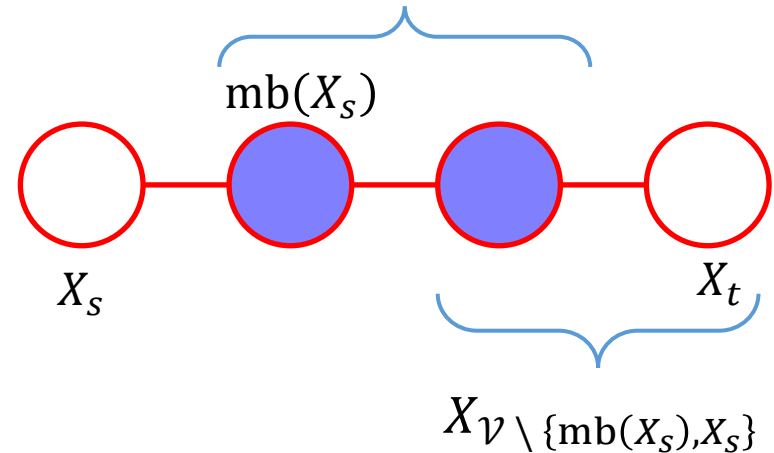
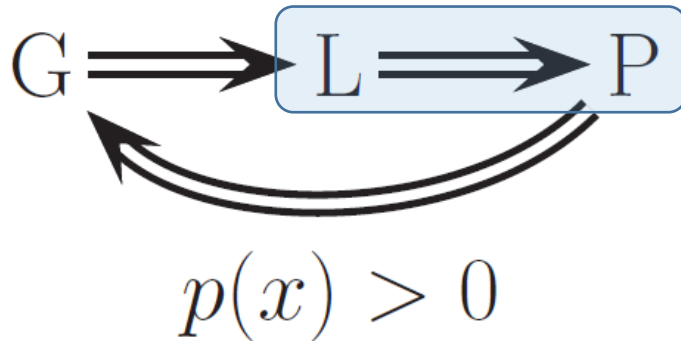
The **global Markov property** implies the **local Markov property**: this is the case when the sets $X_A = X_s$, $X_C = \text{mb}(X_s)$, and $X_B = \mathcal{V} \setminus \{\text{mb}(X_s), X_s\}$.

$$X_A \perp X_B \mid X_C \Rightarrow X_s \perp \mathcal{V} \setminus \{\text{mb}(X_s), X_s\} \mid \text{mb}(X_s)$$

Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Conditional Independence

Local Markov implies pairwise Markov: $X_{\mathcal{V} \setminus \{X_s, X_t\}}$



Proof Sketch :

Given any node X_t that is not adjacent to the node X_s , it follows from **local Markov property** that:

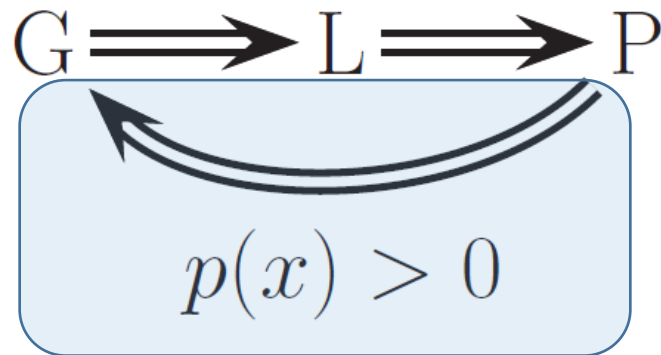
$$X_s \perp X_{\mathcal{V} \setminus \{mb(X_s), X_s\}} \mid mb(X_s)$$

This implies $X_s \perp X_t \mid X_{\mathcal{V} \setminus \{X_s, X_t\}}$, i.e. **pairwise Markov property**.

Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Conditional independence

Pairwise Markov implies global Markov:



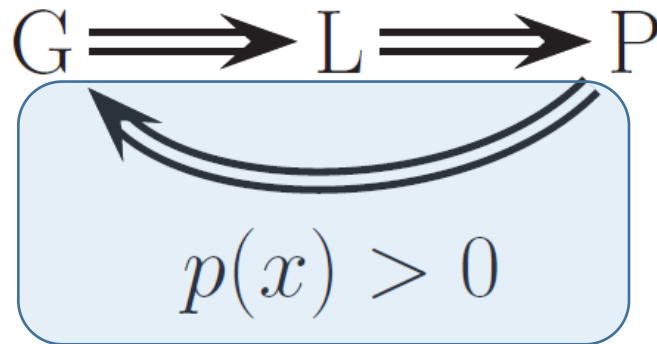
Proof Sketch:

Let $S, A, B, D \subset \mathcal{V}$ be disjoint sets of nodes with S separating A from B in the graph $\mathcal{G}$, where $A \neq \emptyset$ and $B \neq \emptyset$. We will show a sketch of proof that **pairwise Markov** implies **global Markov** using backward induction.

Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Conditional independence

Pairwise Markov implies global Markov:



Proof Sketch :

Let $d = |\mathcal{V}|$, when $|S| = d - 2$:

$A \perp B \mid S$, where $|A| = |B| = 1$

$\Rightarrow$ pairwise Markov

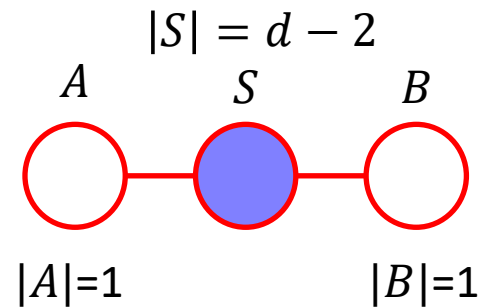
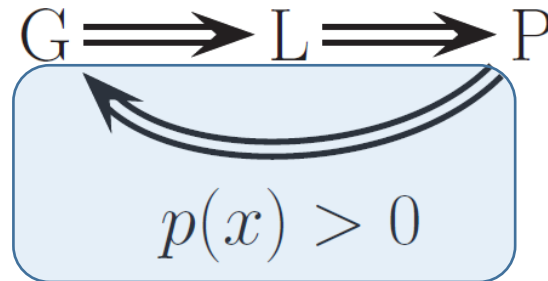


Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Conditional independence

Pairwise Markov implies global Markov:



Proof Sketch :

For $|S| < d - 2$, WLOG, let us assume the set of nodes D is **not connected** to B , where $|D| \geq 1$, $|A| \geq 1$ and $|B| \geq 1$.

We have:

$$A \perp B \mid \{S, D\} \text{ and } B \perp D \mid \{A, S\}$$

Intersection
Lemma



$$B \perp \{A, D\} \mid S \Rightarrow \text{global Markov}$$

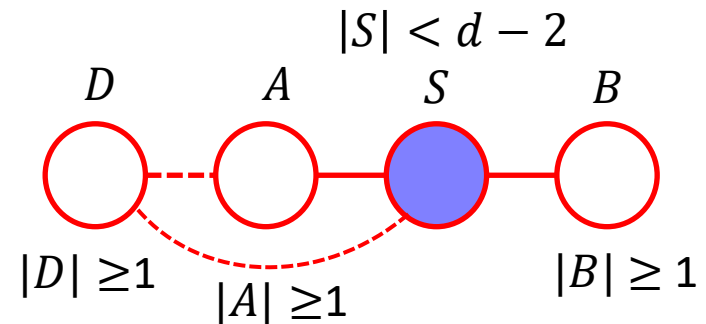
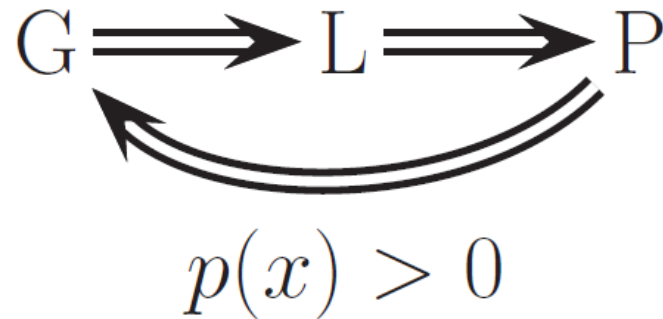


Image Source: "Machine Learning – A Probabilistic Perspective", Kevin Murphy

Conditional independence



- The importance of this result is that it is usually **easier** to empirically **assess pairwise conditional independence**.
- Such pairwise CI statements **can be used to construct a graph** from which global CI statements can be extracted.

Image Source: “Machine Learning – A Probabilistic Perspective”, Kevin Murphy

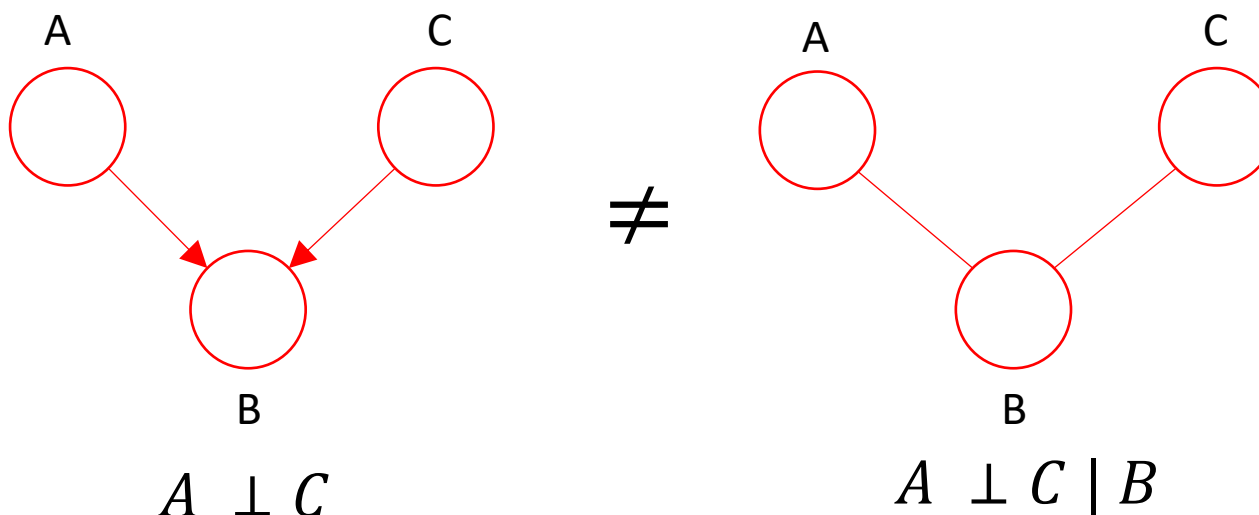
Comparative Semantics

- We have seen that it is **easier** to determine conditional independence using UGMs than DGMs.
- **Question:** Can we determine conditional independence in a DGM using a UGM, or vice versa?
- This is **NOT possible in general!**

Comparative Semantics

- It is tempting to simply convert the DGM to a UGM by **dropping the orientation of the edges**, but this is **not always** correct!

Example:

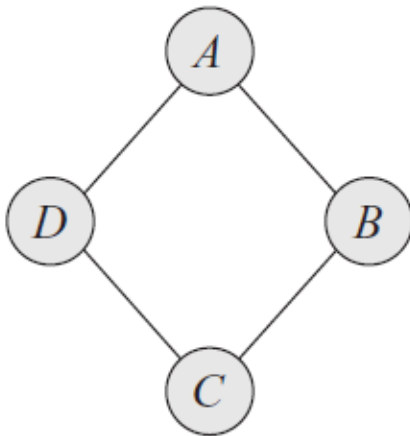


This conditional independence
is **NOT** in the DGM!

Comparative Semantics

- An example of some CI relationships that can be perfectly modeled by a UGM but not a DGM:

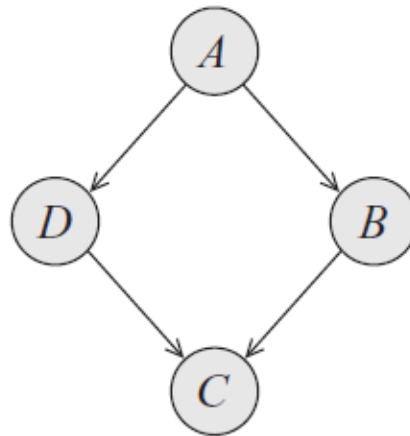
UGM



$$A \perp C \mid \{B, D\}$$

$$B \perp D \mid \{A, C\}$$

Attempt 1

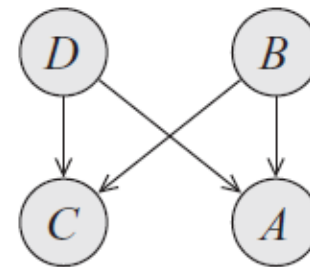


$$\checkmark A \perp C \mid \{B, D\}$$

$$\checkmark B \perp D \mid A$$

$$\times B \perp D \mid \{A, C\}$$

Attempt 2



$$\checkmark A \perp C \mid \{B, D\}$$

$$\checkmark B \perp D$$

$$\times B \perp D \mid \{A, C\}$$

Image source: Koller and Friedman 2009

Parameterization of MRFs

- As in the case of DGMs, we would like to obtain a **local parameterization** for UGMs.
- We have seen earlier that for **DGMs**:
 - Parameterization was based on **local conditional probabilities** of a node and its parents, i.e. $p(x_i|x_{\pi_i})$.
 - Joint probability is a **product of local conditional probabilities** because of the chain rule, i.e.

$$p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i|x_{\pi_i})$$

Parameterization of MRFs

- Difficult to do local parameterization based on conditional probabilities since **no topological ordering** associated with UGMs.
- It turns out that it's better to **abandon conditional probabilities** altogether and **use some functions** instead.

Parameterization of MRFs

- **Lose the ability** to give local probabilistic interpretation to the functions used to represent the joint probability.
- **Retain the ability** the all-important representation of the joint as a product of local functions.

Parameterization of MRFs

How do we decide the domain of the **local functions**?

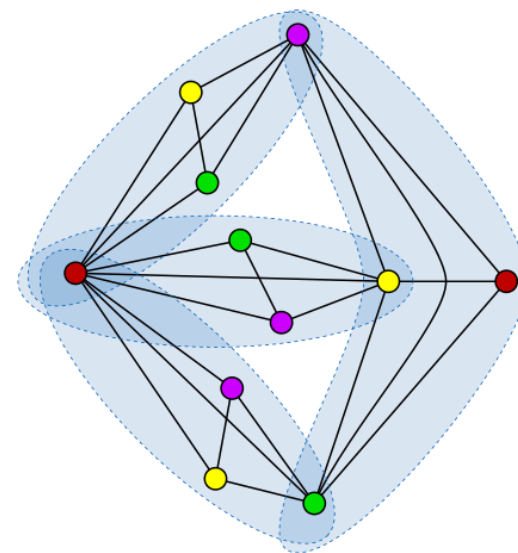
- Recall two nodes X_i and X_j that are not directly linked in an UGM are **conditionally independent** given all other nodes.
- Thus, it must be possible to obtain a factorization of the joint probability that **places X_i and X_j in different factors**.
- This implies that we **cannot have** a local function that depends on both X_i and X_j .

$$p(x_1, \dots, x_N) \neq \psi_1(x_i, x_j, \dots) \dots \psi_m(\dots)$$

Parameterization of MRFs

How do we decide the domain of the **local functions**?

- Our argument thus far suggested that all nodes X_C that belong to a **maximal clique** C in the UGM appear together in a local function $\psi(x_C)$.
- A **clique** of a graph is a fully-connected subset of nodes.
- The **maximal cliques** of a graph are the cliques that cannot be extended to include additional nodes without losing the property of being fully connected.



Conditional independence is impossible for any two nodes in maximal clique!

Image source: [http://wikivisually.com/wiki/Clique_\(graph_theory\)](http://wikivisually.com/wiki/Clique_(graph_theory))

Parameterization of MRFs

Hammersley-Clifford theorem:

A **positive distribution** $p(y) > 0$ satisfies the CI properties of an undirected graph $\mathcal{G}$ iff p can be represented as a **product of factors**, one per maximal clique:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

where

- $\mathcal{C}$ is the set of all the maximal cliques of $\mathcal{G}$
- $\psi_c(\cdot)$ is the **factor** or **potential function** of clique c
- θ is the parameter of the factors $\psi_c(\cdot)$ for $c \in \mathcal{C}$
- $Z(\theta)$ is the **partition function**

Parameterization of MRFs

Hammersley-Clifford theorem:

$Z(\theta)$ is the **partition function** given by:

$$Z(\theta) \triangleq \sum_{\mathbf{y}} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c | \theta_c)$$

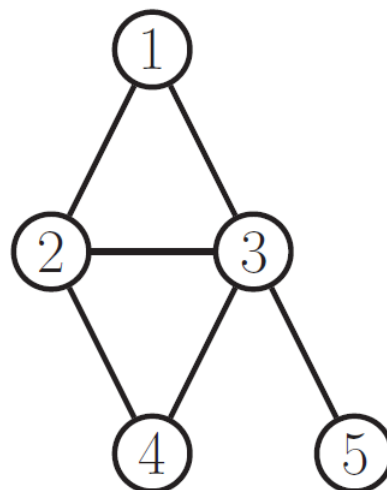
Since $\psi(\cdot)$ can be any arbitrary positive function, the partition function $Z(\theta)$ ensures the **overall distribution sums to 1**.

Parameterization of MRFs

Example:

Assume: $y_i \in \{0,1\}$

y_2	y_3	y_4	ψ_{234}
0	0	0	a_1
0	0	1	a_2
0	1	0	a_3
0	1	1	a_4
1	0	0	a_5
1	0	1	a_6
1	1	0	a_7
1	1	1	a_8



y_1	y_2	y_3	ψ_{123}
0	0	0	b_1
0	0	1	b_2
0	1	0	b_3
0	1	1	b_4
1	0	0	b_5
1	0	1	b_6
1	1	0	b_7
1	1	1	b_8

y_3	y_5	ψ_{35}
0	0	c_1
0	1	c_2
1	0	c_3
1	1	c_4

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \psi_{35}(y_3, y_5)$$

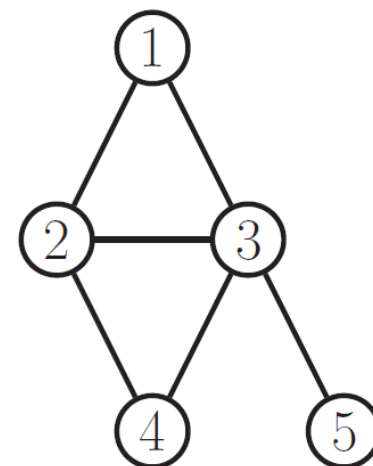
where

$$Z = \sum_{\mathbf{y}} \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \psi_{35}(y_3, y_5)$$

“An introduction to probabilistic graphical models”, Michael I. Jordan, 2002

Pairwise Parameterization of MRFs

- Parameterization of MRFs is **not unique**!!!
- We are **free to relax** the parameterization to the edges of the graph, rather than the maximal cliques.
- This is **pairwise MRF**, and is widely used due to its simplicity, although it is not as general.



Example:

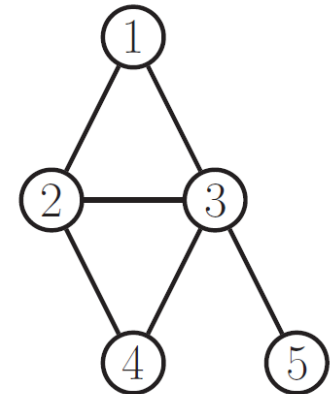
$$\begin{aligned} p(\mathbf{y}|\boldsymbol{\theta}) &\propto \psi_{12}(y_1, y_2)\psi_{13}(y_1, y_3)\psi_{23}(y_2, y_3)\psi_{24}(y_2, y_4)\psi_{34}(y_3, y_4)\psi_{35}(y_3, y_5) \\ &\propto \prod_{s \sim t} \psi_{st}(y_s, y_t) \end{aligned}$$

Canonical Parameterization of MRFs

- Another way is to define the parameterization over **all cliques** in the graph, i.e., **canonical parameterization**.
- **Uniform prior** can be assumed on any potential function.

Example: Assume: $y_i \in \{0,1\}$

Uniform Prior on pairwise potentials (**optional**)



$$p(\mathbf{y} | \theta) \propto \psi_1(y_1)\psi_2(y_2)\psi_3(y_3)\psi_4(y_4)\psi_5(y_5) \\
\cancel{\psi_{12}(y_1, y_2)}\cancel{\psi_{23}(y_2, y_3)}\cancel{\psi_{13}(y_1, y_3)} \\
\cancel{\psi_{24}(y_2, y_4)}\cancel{\psi_{34}(y_3, y_4)}\cancel{\psi_{35}(y_3, y_5)} \\
\psi_{123}(y_1, y_2, y_3)\psi_{234}(y_2, y_3, y_4)$$

$$= \psi_1(y_1)\psi_2(y_2)\psi_3(y_3)\psi_4(y_4)\psi_5(y_5) \\
\psi_{123}(y_1, y_2, y_3)\psi_{234}(y_2, y_3, y_4)$$

y_i	y_j	ψ_{ij}
0	0	1
0	1	1
1	0	1
1	1	1

Gibbs Distribution

- There is a deep connection between UGMs and **statistical physics**.
- In particular, there is a model known as the **Gibbs distribution**, which can be written as follows:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp\left(-\sum_c E(\mathbf{y}_c|\boldsymbol{\theta}_c)\right)$$

- $E(\mathbf{y}_c) > 0$ is the **energy** associated with the variables in clique c .

Gibbs Distribution

- We can convert the Gibbs distribution to a UGM by defining:

$$\psi_c(\mathbf{y}_c | \boldsymbol{\theta}_c) = \exp(-E(\mathbf{y}_c | \boldsymbol{\theta}_c))$$

$$\Rightarrow p(\mathbf{y} | \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp(-\sum_c E(\mathbf{y}_c | \boldsymbol{\theta}_c)) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c | \boldsymbol{\theta}_c)$$

- We see that **high probability states** correspond to **low energy configurations**.
- Also known as **energy-based models**, hence the term **“potential function”** for $\psi_c(\cdot)$.

Log-Linear Potential Functions

- Potentials represent the **relative “compatibility”** between the different assignments to the random variables.
- A general approach is to define the **log potentials** as a **linear function of the parameters** $\theta_c \in \mathbb{R}^M$:

$$-E(\mathbf{y}_c) = \log \psi_c(\mathbf{y}_c) \triangleq \phi_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c$$

- $\phi_c: \mathbf{y}_c \mapsto \mathbb{R}^M$ is a **feature vector** derived from the values of the variables $\mathbf{y}_c$.

Log-Linear Potential Functions

- The resulting **log probability** has the form:

$$\log p(\mathbf{y}|\boldsymbol{\theta}) = \sum_c \phi_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c - \log Z(\boldsymbol{\theta})$$

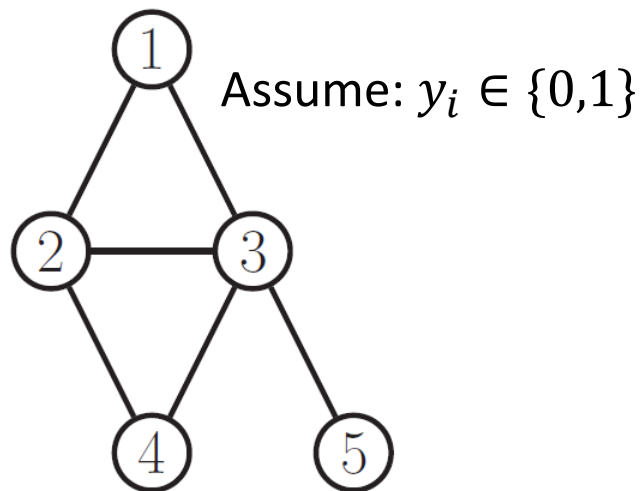
- This is also known as a **maximum entropy** or a **log-linear** model.

Log-Linear Potential Functions

Example: $\log \psi_c(\mathbf{y}_c) \triangleq \phi_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c$

$$\Rightarrow \psi_c(\mathbf{y}_c) = \exp\{\phi_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c\}$$

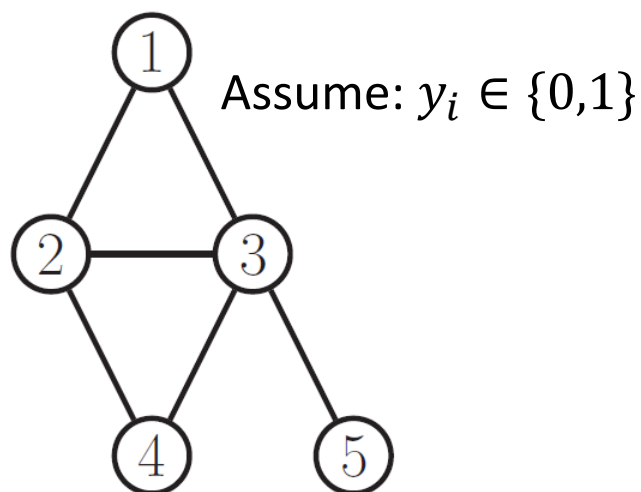
Let's look at $\mathbf{y}_c = \{y_1, y_2, y_3\}$, we set $M = |c|$, such that $\boldsymbol{\theta}_c \in \mathbb{R}^M$ and $\phi_c: \mathbf{y}_c \mapsto \mathbb{R}^M$.



y_1	y_2	y_3	ψ_{123}
0	0	0	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 0, y_3 = 0)^T \boldsymbol{\theta}_c\}$
0	0	1	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 0, y_3 = 1)^T \boldsymbol{\theta}_c\}$
0	1	0	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 1, y_3 = 0)^T \boldsymbol{\theta}_c\}$
0	1	1	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 1, y_3 = 1)^T \boldsymbol{\theta}_c\}$
1	0	0	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 0, y_3 = 0)^T \boldsymbol{\theta}_c\}$
1	0	1	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 0, y_3 = 1)^T \boldsymbol{\theta}_c\}$
1	1	0	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 1, y_3 = 0)^T \boldsymbol{\theta}_c\}$
1	1	1	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 1, y_3 = 1)^T \boldsymbol{\theta}_c\}$

Log-Linear Potential Functions

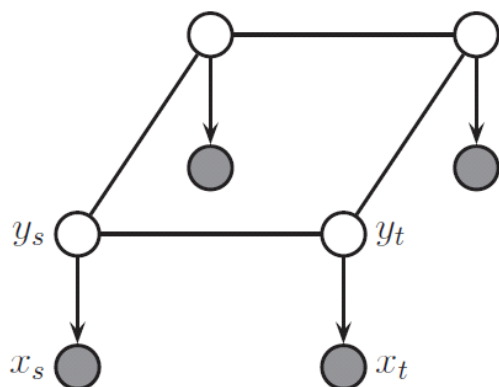
- In this case, we set $M = 3$ and $\phi_c: y_c \mapsto [y_1, y_2, y_3]^\top$, we **only need 3 parameters** $\theta_c = [\theta_{c1}, \theta_{c2}, \theta_{c3}]$ instead of 2^3 parameters.



y_1	y_2	y_3	ψ_{123}
0	0	0	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 0, y_3 = 0)^\top \theta_c\}$
0	0	1	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 0, y_3 = 1)^\top \theta_c\}$
0	1	0	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 1, y_3 = 0)^\top \theta_c\}$
0	1	1	$\propto \exp\{\phi_c(y_1 = 0, y_2 = 1, y_3 = 1)^\top \theta_c\}$
1	0	0	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 0, y_3 = 0)^\top \theta_c\}$
1	0	1	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 0, y_3 = 1)^\top \theta_c\}$
1	1	0	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 1, y_3 = 0)^\top \theta_c\}$
1	1	1	$\propto \exp\{\phi_c(y_1 = 1, y_2 = 1, y_3 = 1)^\top \theta_c\}$

Combining UGM and DGM

Example: Ising and Potts Models



Depth Map from Stereo Images

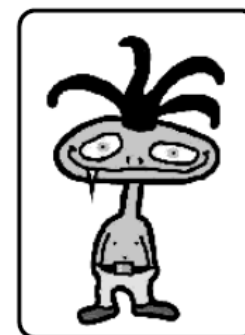
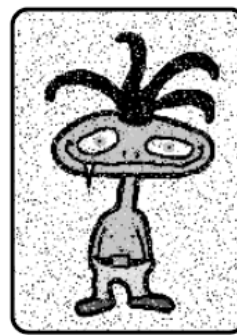


Observed Variables
 $x \in \{1, \dots, L\}$

Latent Variables
 $y \in \{1 \dots L\}$

$$\begin{aligned}
 p(y, x | J, \theta) &= p(\mathbf{y} | J) \prod_t p(x_t | y_t, \theta) \\
 &= \underbrace{\left[\frac{1}{Z(J)} \prod_{s \sim t} \psi(y_s, y_t; J) \right]}_{\text{Pairwise potential}} \underbrace{\prod_t p(x_t | y_t, \theta)}_{\text{Unary potential}}
 \end{aligned}$$

Binary Image Denoising



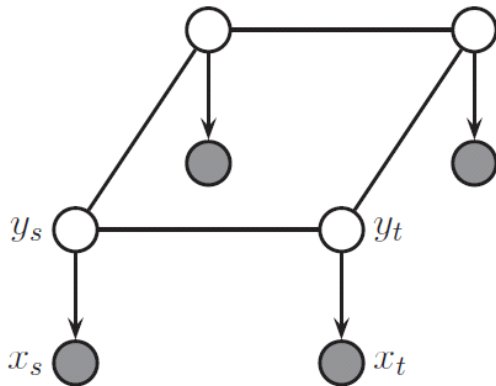
Observed Variables
 $x \in \{0, 1\}$

Latent Variables
 $y \in \{0, 1\}$

Image Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Combining UGM and DGM

Example: Ising and Potts Models



$$\begin{aligned} p(y, x | J, \theta) &= p(y | J) \prod_t p(x_t | y_t, \theta) \\ &= \left[\frac{1}{Z(J)} \underbrace{\prod_{s \sim t} \psi(y_s, y_t; J)}_{\text{Pairwise potential}} \right] \underbrace{\prod_t p(x_t | y_t, \theta)}_{\text{Unary potential}} \end{aligned}$$

Ising Model:

$$y_i \in \{0, 1\}, \quad x_i \in \{0, 1\}$$

$$E(y_s, y_t; J) = J |y_s - y_t|, \\ J > 0$$

Potts Model:

$$y_i \in \{1, \dots, L\}, \quad x_i \in \{1, \dots, L\}$$

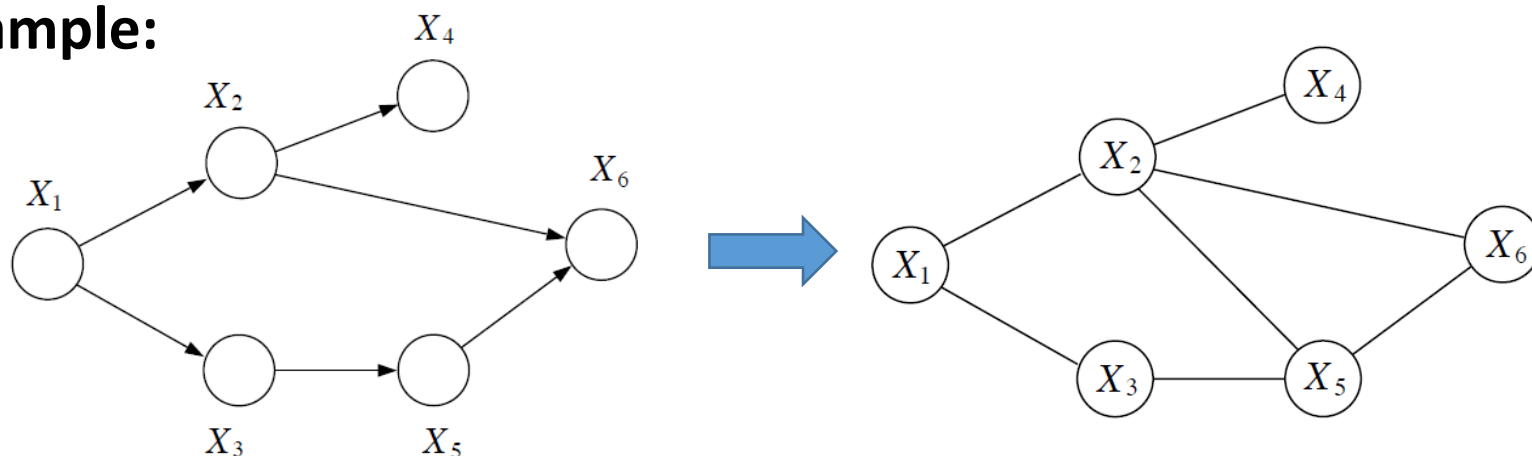
$$E(y_s, y_t; J) = J \min(|y_s - y_t|, 1), \\ J > 0$$

Image Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Representing Potential Functions

- We can also use **local conditional probabilities** from a DGM to represent the potential functions in a UGM.

Example:



$$p(x) = \frac{1}{Z} \underbrace{\varphi_{12}(x_1, x_2)}_{p(x_2|x_1)} \underbrace{\varphi_{13}(x_1, x_3)}_{p(x_3|x_1)} \underbrace{\varphi_{24}(x_2, x_4)}_{p(x_4|x_2)} \underbrace{\varphi_{35}(x_3, x_5)}_{p(x_5|x_3)} \underbrace{\varphi_{256}(x_2, x_5, x_6)}_{p(x_6|x_2, x_5)}$$

$$Z = \sum_x \varphi_{12}(x_1, x_2) \varphi_{13}(x_1, x_3) \varphi_{24}(x_2, x_4) \varphi_{35}(x_3, x_5) \varphi_{256}(x_2, x_5, x_6) = \frac{1}{p(x_1)}$$

Image source: "An introduction to probabilistic graphical models", Michael I. Jordan, 2002.

Moralization

- A DGM can be converted into a UGM by “marrying” the unmarried parents of a node, i.e. **moralization**.
- This process **preserves the joint distribution**, but **conditional independence is lost**!
- Moralization is important for **exact inference** (next lectures).

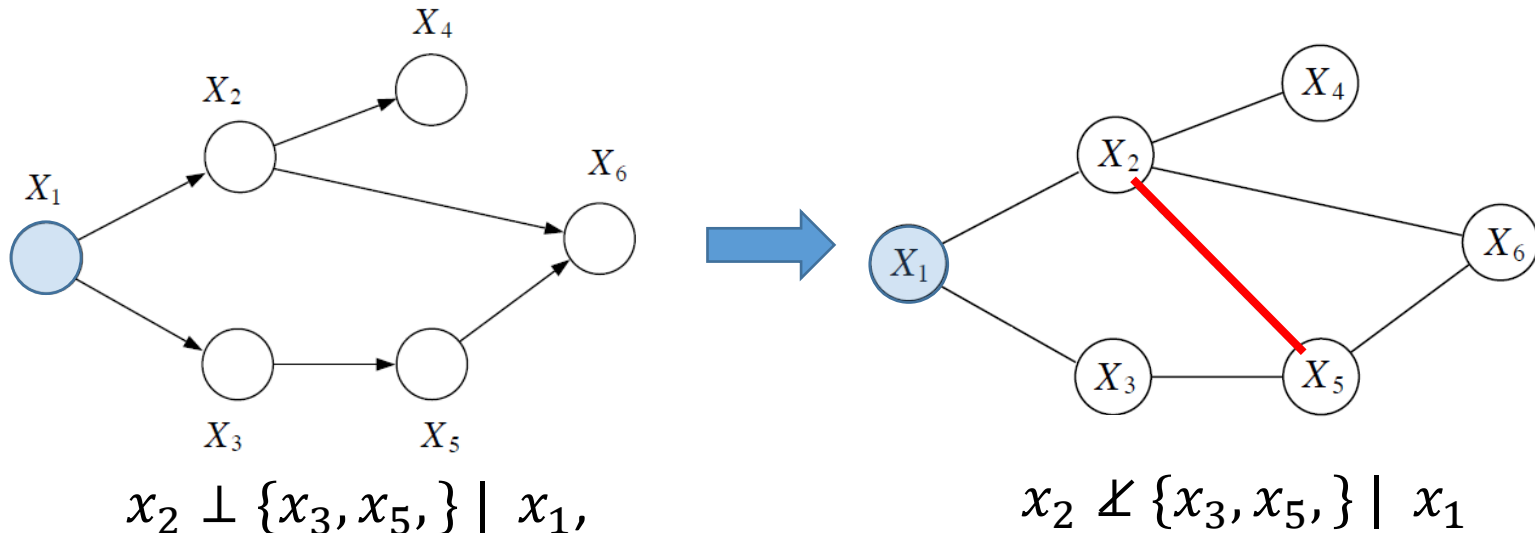


Image source: “An introduction to probabilistic graphical models”, Michael I. Jordan, 2002.

Discriminative Vs Generative Models

- **Generative models:** Approaches that explicitly or implicitly model the distribution of inputs and outputs.
- Sampling from the distribution it is possible to generate synthetic data points in the input space.

Likelihood: $p(\mathbf{x}|\mathcal{C}_k)$

- **Discriminative models:** Approaches that model the posterior probabilities directly.

Posterior: $p(\mathcal{C}_k|\mathbf{x})$

Conditional Random Fields

- A **CRF** or **discriminative random field**, is just a version of an MRF where all the clique potentials are **conditioned on all observed random variables X** :

$$p(\mathbf{y}|\mathbf{x}, \mathbf{w}) = \frac{1}{Z(\mathbf{x}, \mathbf{w})} \prod_c \psi_c(\mathbf{y}_c|\mathbf{x}, \mathbf{w})$$

- We will usually assume a **log-linear representation** of the potentials:

$$\psi_c(\mathbf{y}_c|\mathbf{x}, \mathbf{w}) = \exp(\mathbf{w}_c^T \phi(\mathbf{x}, \mathbf{y}_c))$$

- where $\phi(\mathbf{y}_c, \mathbf{x})$ is a **feature vector** derived from the **latent** random variables Y_c and the **observed** random variables X , and $\mathbf{w}_c$ **is the parameters**.

CRF vs MRF

Advantages:

1. **No need to “waste resources”** modeling things that we always observe.

Focus our attention on **modeling what we care about**, i.e. the distribution of latent given the observed random variables.

2. We can make the potentials (or factors) of the model be **data-dependent**.

e.g. in natural language processing problems, we can make the **latent labels depend on observations** of the sentence, such as which language it is written in.

CRF vs MRF

Disadvantage:

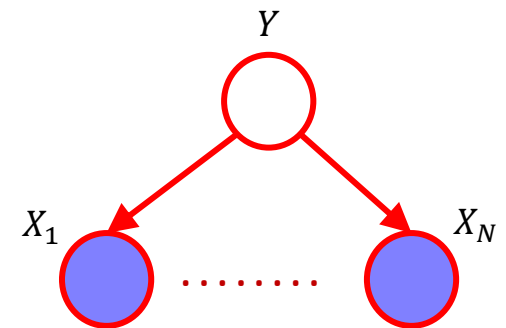
1. Not generative, this means we **cannot generate new samples**.
2. Learning is **slower** (more detail in the coming lectures).

CRF: Logistic Regression

- Naives Bayes classifier we have seen previously **ignores dependencies** among the observations $X_1, \dots, X_N$.

Joint distribution:

$$p(x_1, \dots, x_N, Y) = \prod_{n=1}^N p(x_n | Y)$$



Predict with Bayes rule:

$$p(Y = 1 | x_1, \dots, x_N) = \frac{p(y = 1) \prod_{n=1}^N p(x_n | Y = 1)}{\sum_{y=\{0,1\}} p(Y = y) \prod_{n=1}^N p(x_n | Y = y)}$$

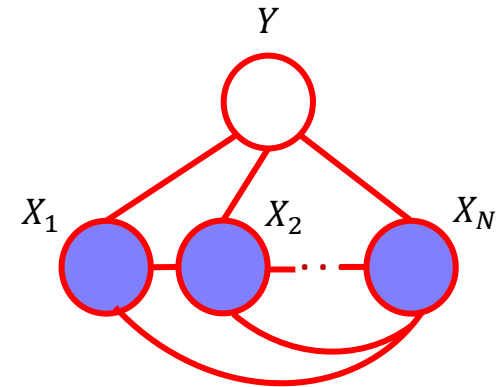
CRF: Logistic Regression

- In CRF, we **model the full dependencies** among the observations $X_1, \dots, X_N$.

$$p(\mathbf{y}|\mathbf{x}, \mathbf{w}) = \frac{1}{Z(\mathbf{x}, \mathbf{w})} \psi(\mathbf{y} | \mathbf{x}, \mathbf{w})$$

$$= \frac{\exp(\phi(\mathbf{y}, \mathbf{x})^\top \mathbf{w})}{\sum_{\mathbf{y}} \exp(\phi(\mathbf{y}, \mathbf{x})^\top \mathbf{w})}$$

$$= \frac{\exp(\sum_n \phi_n(\mathbf{y}, \mathbf{x}) w_n)}{\sum_{\mathbf{y}} \exp(\sum_n \phi_n(\mathbf{y}, \mathbf{x}) w_n)} \quad (\text{Log-linear model})$$

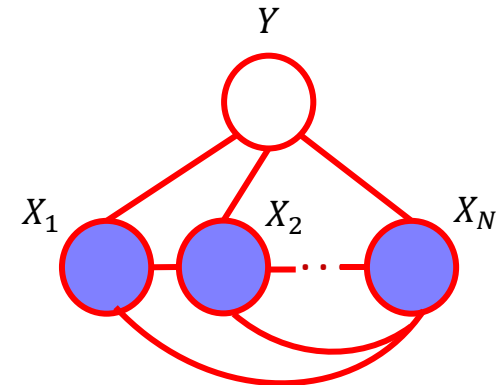


CRF: Logistic Regression

- In CRF, we **model the full dependencies** among the observations $X_1, \dots, X_N$.

For $y \in \{0,1\}$, e.g., email spam classifier, we can define:

$$\begin{aligned}\phi_n(y = 1, \mathbf{x})w_n &= \mathbb{I}(y^+)x_nw_n^+, \text{ and} \\ \phi_n(y = 0, \mathbf{x})w_n &= \mathbb{I}(y^-)x_nw_n^-.\end{aligned}$$



We get:

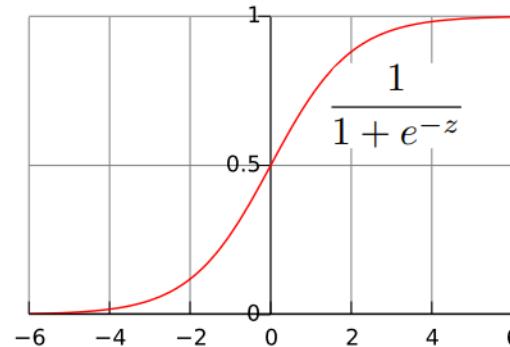
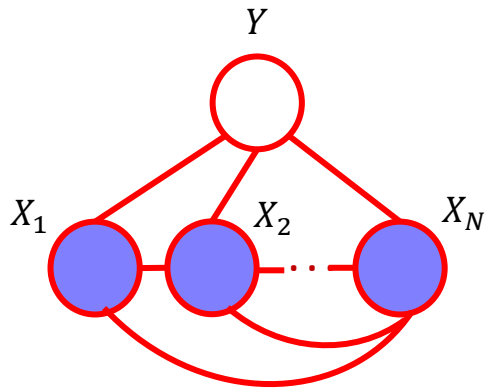
$$p(y^+ | \mathbf{x}) = \frac{\exp(\mathbf{w}^{+\top} \mathbf{x})}{\exp \mathbf{w}^{+\top} \mathbf{x} + \exp \mathbf{w}^{-\top} \mathbf{x}} = \frac{1}{1 + \exp(-\mathbf{w}'^{\top} \mathbf{x})}$$

Logistic Regression!

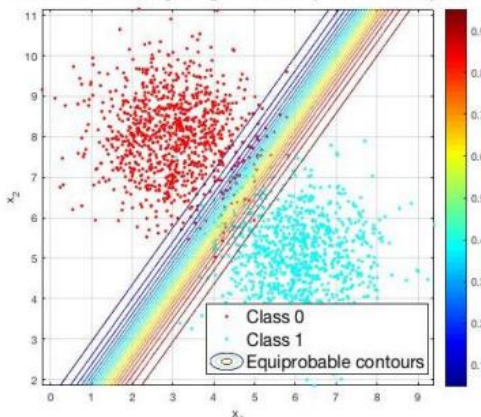
where $\mathbf{w}' = \mathbf{w}^+ - \mathbf{w}^-$.

Note: there's just 1 set of parameters, i.e., $\mathbf{w}'$, we don't care about modeling the respective $\mathbf{w}^+$ and $\mathbf{w}^-$.

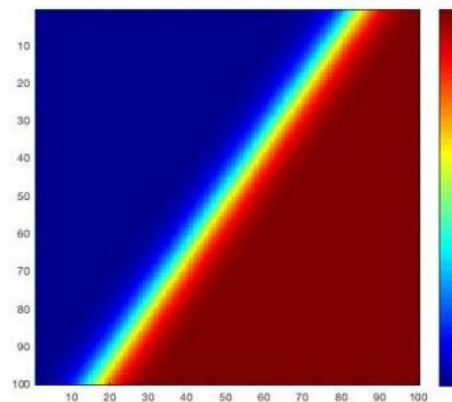
CRF: Logistic Regression



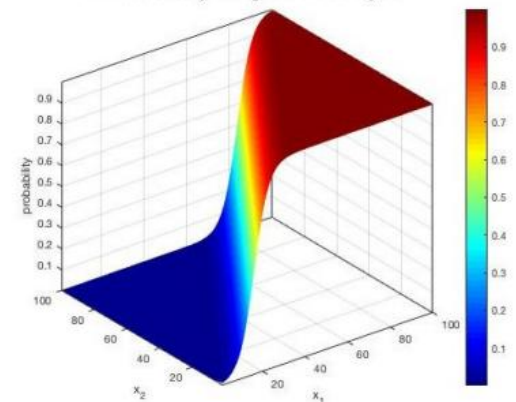
Contours of equal probability defined by θ



Probability map defined by θ



Probability map defined by θ

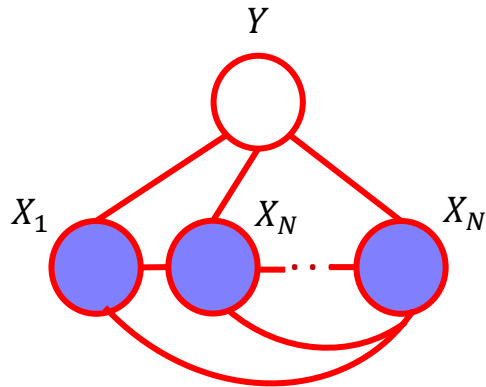


- **Decision boundary:** $p(y = 1 \mid \mathbf{x}, \mathbf{w}) > 0.5$ is linear in $\mathbf{x}$.

Image source:

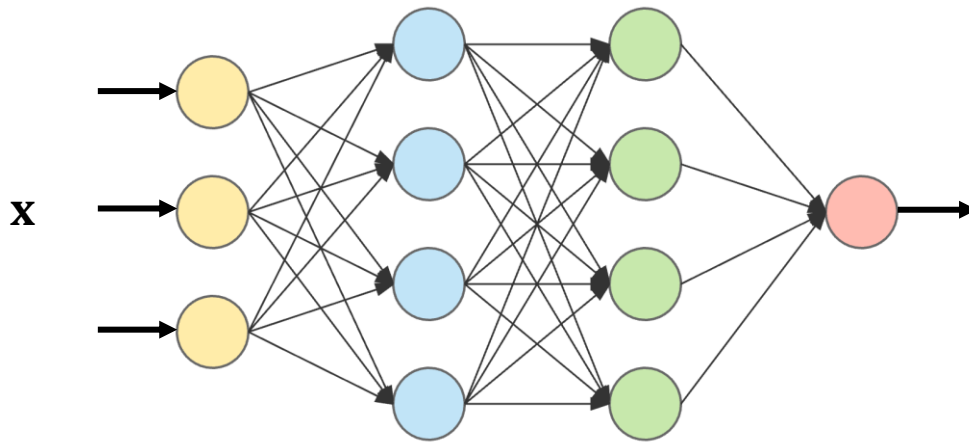
https://deepgenerativemodels.github.io/assets/slides/cs236_lecture2.pdf

Classification Deep Network



$$p(y^+ | \mathbf{x}) = \frac{1}{1 + \exp \mathbf{w}'^T \mathbf{x}}$$

- The CRF becomes a deep network when $\mathbf{w}'^T \mathbf{x}$ is a **nonlinear function** $f(\mathbf{x}; \mathbf{w})$!



$\mathbf{w}$: weight parameters

$$p(y^+ | \mathbf{x}) = \frac{1}{1 + \exp f(\mathbf{x}; \mathbf{w})}$$

$$= \sigma(f(\mathbf{x}; \mathbf{w})),$$

where $\sigma(\cdot)$ is the sigmoid function.

Note: $\sigma(\cdot)$ is the **softmax function** for multiclass classification.

CRF: Linear Chain CRF

Models for sequential data

Hidden Markov model:

$$p(\mathbf{x}, \mathbf{y} | \mathbf{w}) = \prod_{t=1}^T p(y_t | y_{t-1}, \mathbf{w}) \underbrace{p(\mathbf{x}_t | y_t, \mathbf{w})}_{\text{Likelihood, i.e. generative}}$$

Likelihood, i.e. generative

Chain structure MRF:

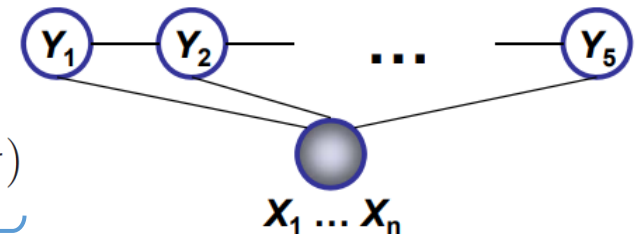
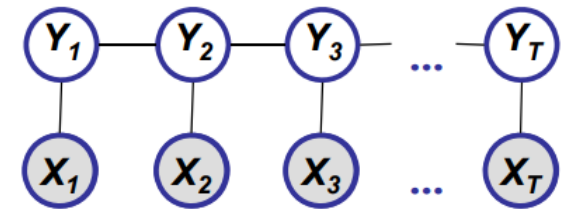
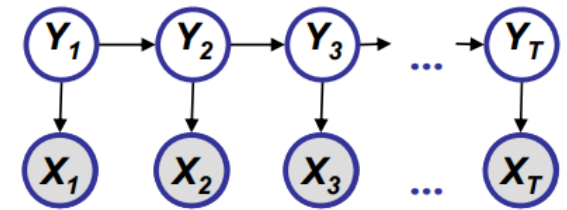
$$p(\mathbf{x}, \mathbf{y} | \mathbf{w}) = \prod_{t=1}^T \underbrace{\psi(y_t; \mathbf{x}_t, \mathbf{w})}_{\text{Likelihood, i.e. generative}} \prod_{t=1}^{T-1} \psi(y_t, y_{t+1}; \mathbf{w})$$

Likelihood, i.e. generative

Chain structure CRF:

$$p(\mathbf{y} | \mathbf{x}, \mathbf{w}) = \frac{1}{Z(\mathbf{x}, \mathbf{w})} \prod_{t=1}^T \underbrace{\psi(y_t; \mathbf{x}, \mathbf{w})}_{\text{Posterior, i.e. discriminative}} \prod_{t=1}^{T-1} \underbrace{\psi(y_t, y_{t+1}; \mathbf{x}, \mathbf{w})}_{\text{Posterior, i.e. discriminative}}$$

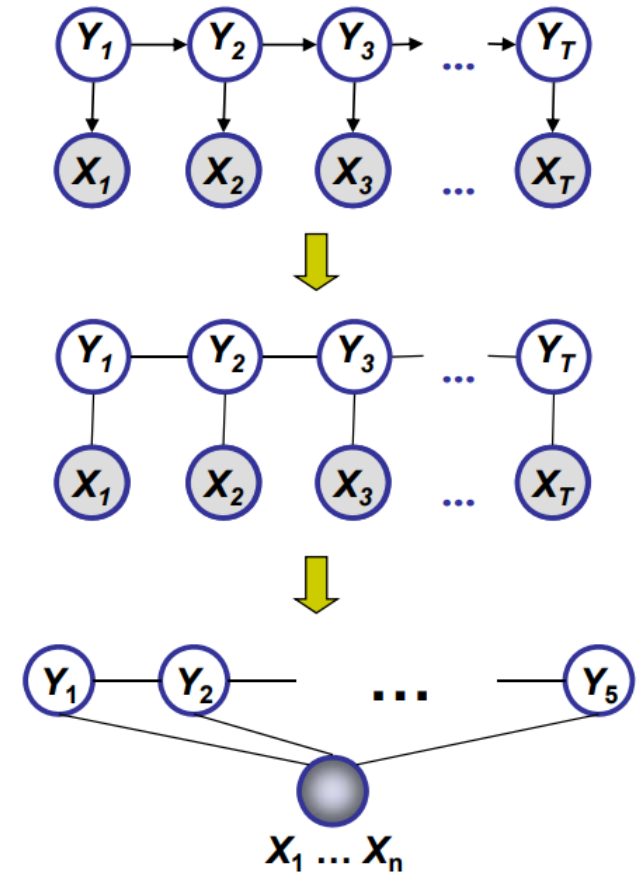
Posterior, i.e. discriminative



CRF: Linear Chain CRF

Models for sequential data

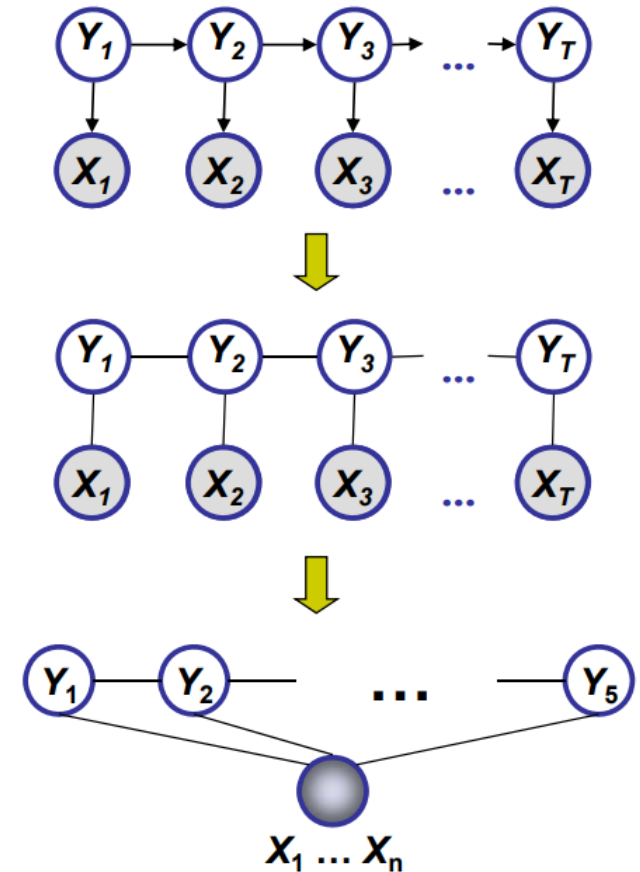
- HMM and MRF suffer from the **label bias problem**.
- Local features at time t do not influence states prior to time t .
- X_t is **d-separated** from all other nodes at Y_t thus blocking the information flow.



CRF: Linear Chain CRF

Models for sequential data

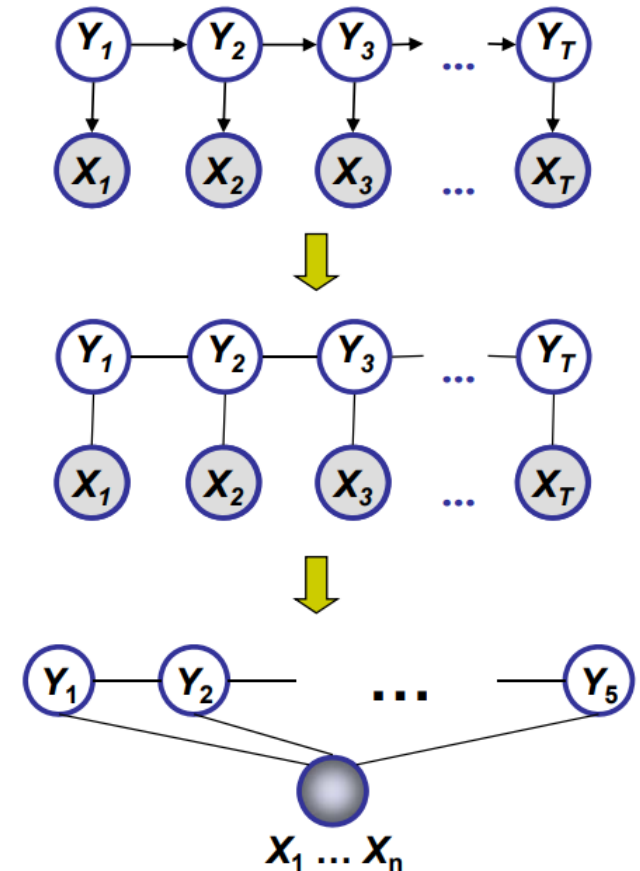
- Consider the **part of speech (POS) tagging** task.
- Suppose we see the word “banks”.
- This could be a **verb** (as in “he banks at DBS”), or a **noun** (as in “the river banks were overflowing”).
- Locally** the POS tag for the word is **ambiguous**.



CRF: Linear Chain CRF

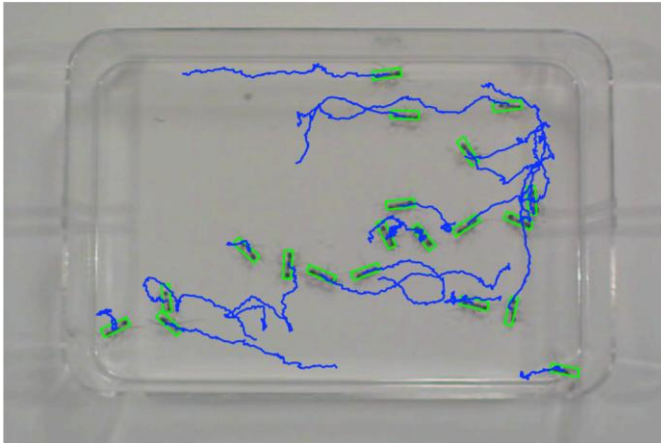
Models for sequential data

- Suppose that later in the sentence, we see the word “fishing”.
- This gives us enough context to infer that the sense of “banks” is “river banks”.
- However, in HMM and MRF the “fishing” evidence is **d-separated**.
- Problem is alleviated in CRF.



Example of MRF

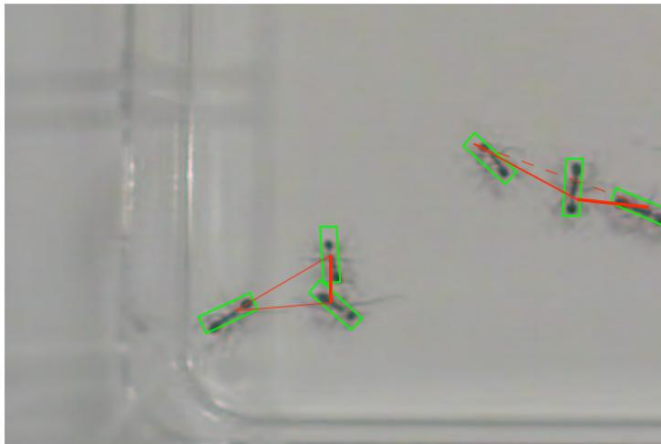
- Objective: Track every ant in the video



$$P(X_t|Z^t) = kP(Z_t|X_t) \int_{X_{t-1}} P(X_t|X_{t-1})P(X_{t-1}|Z^{t-1})$$

x_t : current locations of the ants

z_t : tracklets output (green boxes) from an ant detector



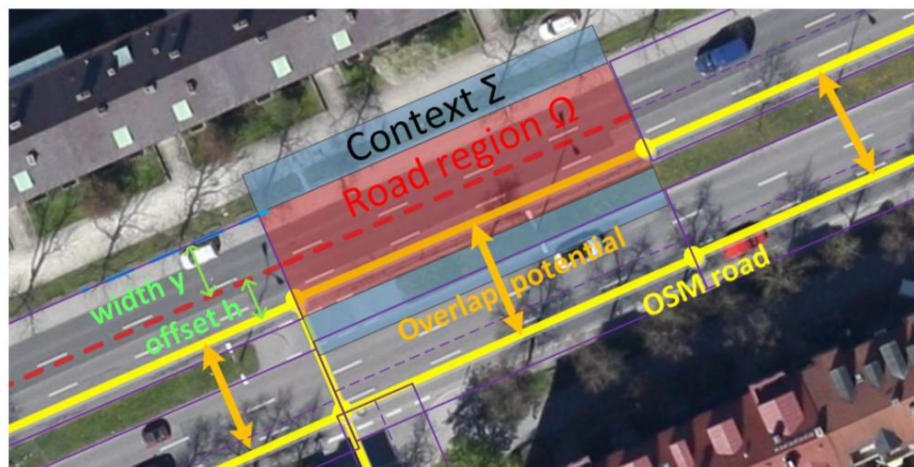
$$P(X_t|X_{t-1}) \propto \prod_i P(X_{it}|X_{i(t-1)}) \prod_{ij \in E} \psi(X_{it}, X_{jt})$$

MRF used to model interactions between the ants.

Zia Khan, Tucker Balch, and Frank Dellaert, **An MCMC-based Particle Filter for Tracking Multiple Interacting Targets**, ECCV 2014

Example of CRF

- **Objective:** To infer the location of the road-segment centerlines as well as their width.



Road classifier: ϕ_{road}

Appearance: $\phi_{ap} = [\phi_{edge}, \phi_{hom}, \phi_{context}]$

Distance to edges: ϕ_{edge}

Homogeneity of the region: ϕ_{hom}

Appearance context: $\phi_{context}$

$$\{\mathbf{h}^*, \mathbf{y}^*\} = \operatorname{argmin}_{\mathbf{h}, \mathbf{y}} E(\mathbf{h}, \mathbf{y})$$

$$\begin{aligned} E(\mathbf{h}, \mathbf{y}) = & \sum_{j=1}^L \sum_{i=1}^{l_j} \mathbf{w}_{road}^T \phi_{road}(h_i^j, y_i^j, \mathbf{x}) \\ & + \sum_{j=1}^L \sum_{i=1}^{l_j} \mathbf{w}_{ap}^T \phi_{ap}(h_i^j, y_i^j, \mathbf{x}) + \sum_{j=1}^L \sum_{i=1}^{l_j} \mathbf{w}_{car}^T \phi_{car}(h_i^j, y_i^j, \mathbf{x}) \\ & + \sum_{j=1}^L \sum_{i=1}^{l_j-1} \mathbf{w}_{sm}^T \phi_{sm}(h_i^j, y_i^j, h_{i+1}^j, y_{i+1}^j) \\ & + \sum_{i,j,k,m \in P} \phi_{ol}(h_i^j, y_i^j, h_k^m, y_k^m) \end{aligned}$$

Smoothness: ϕ_{sm}

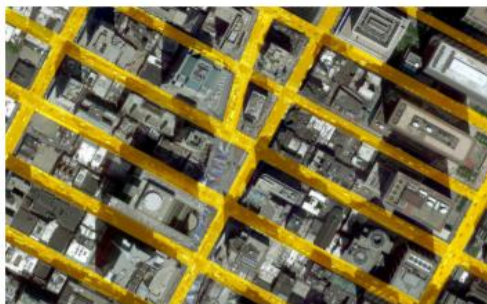
Overlap: ϕ_{ol}

G. Mattyus, S. Wang, S. Fidler and R. Urtasun, **Enhancing World Maps by Parsing Aerial Images**, ICCV 2015

Example of CRF



(Toronto: Pearson Airport)



(NYC: Times square)



(Nairobi, Kenya)*



(Manila, Philippines)



(Mexico City)



(Kyoto: Kinkakuji)



(Sydney: At Harbour bridge)



(St. Moritz, Switzerland)



(Manaus, Brazil)*

G. Mattyus, S. Wang, S. Fidler and R. Urtasun, **Enhancing World Maps by Parsing Aerial Images**, ICCV 2015

Summary

- You have learned how to:
 1. Derive the five rules of conditional independence: **Symmetry**, **Decomposition**, **Weak Union**, **Contraction**, **Intersection**.
 2. Use **I-Map** to explain the correctness of graphical models for conditional independence.
 3. Explain the concepts of **Markov properties (global, local and pairwise)** and use it to find all conditional independences in an UGM.
 4. Use **clique potential functions** to parameterize a Markov Random Field, i.e. to represent the joint distribution with clique potential functions.
 5. Describe the differences and similarities between a **Markov Random Field** and **Conditional Random Field**.